

TRUE Simulation Experiment Report

P0/P1 Refactored Round

Trust-Reliability Unified Engineering
Multi-Agent Collaboration Simulation

100 runs x 200 rounds x 4 scenarios x 7 groups
Fixed task-flow pairing | Bootstrap CIs | Bonferroni correction

Executive Summary

This report presents the P0/P1 refactored experimental results for the TRUE (Trust-Reliability Unified Engineering) multi-agent collaboration framework. All groups within each Monte Carlo run face the identical task sequence, ensuring that outcome differences are attributable to selection mechanisms rather than task luck.

Key changes: (1) Fixed task-flow pairing; (2) Removed group-specific observation manipulation for A8; (3) Renamed MOO to TTB (Trust-Targeted Baseline) and added a proper MOO baseline using weighted Tchebycheff; (4) Added ablation variants TRUE-C, TRUE-E, TRUE-N; (5) Wilcoxon signed-rank test + bootstrap 95% CIs + Bonferroni correction.

Scenario Definitions

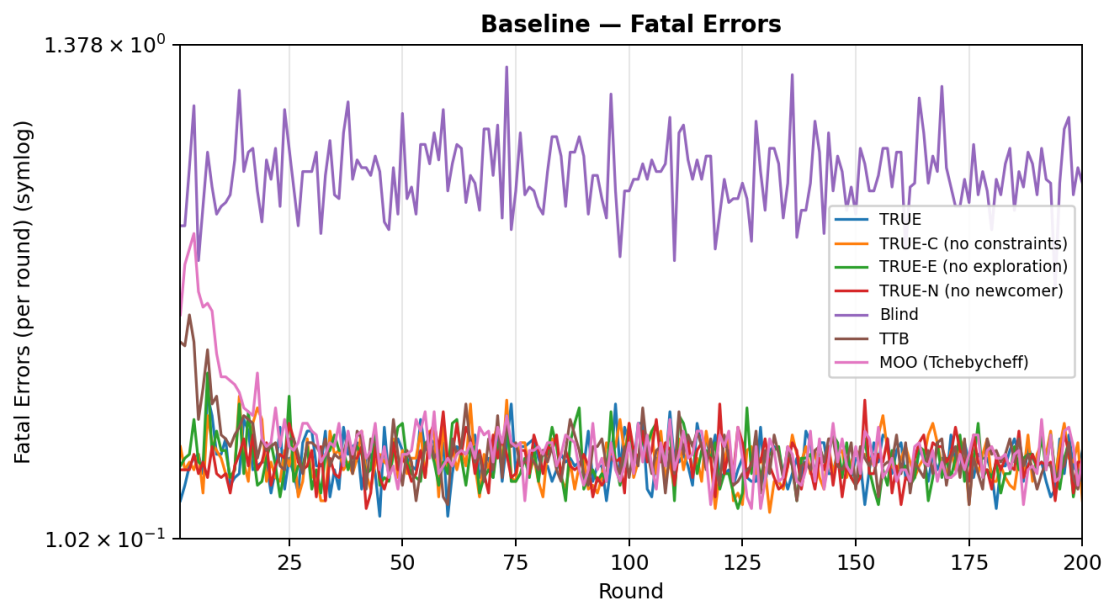
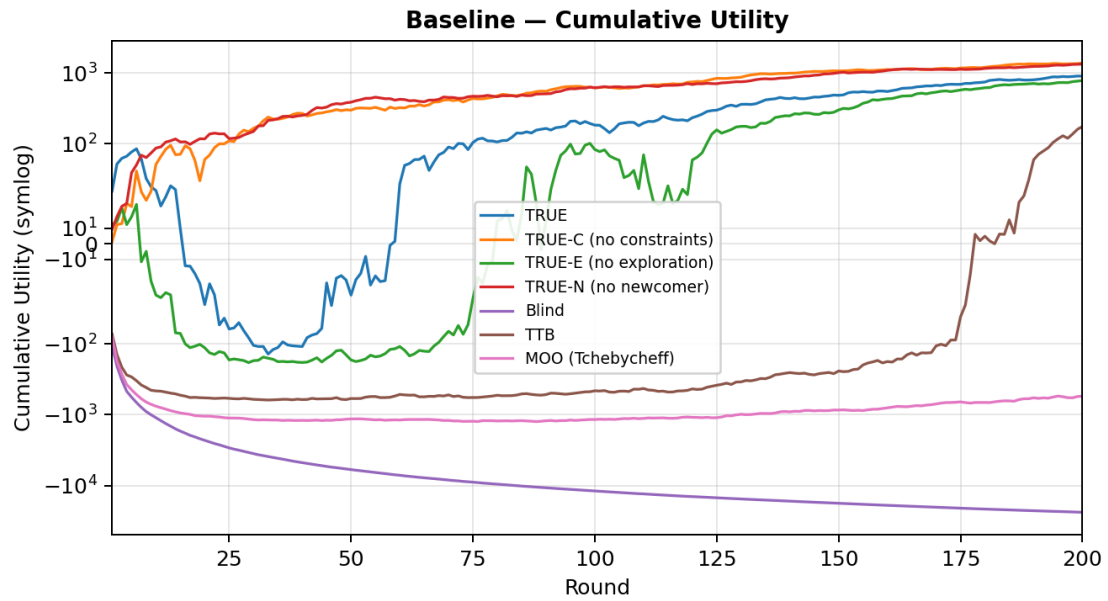
Scenario	Description
Baseline	Standard engineering collaboration with moderate observation noise.
Safety-Critical	High-stakes safety environment with harsher penalties and stronger error propagation.
Observation-Manipulated	Observation-contaminated environment where surface quality diverges from true quality.
Utility-Trust-Misalignment	Local phases where trusted incumbents underperform and low-history entities hold latent value.

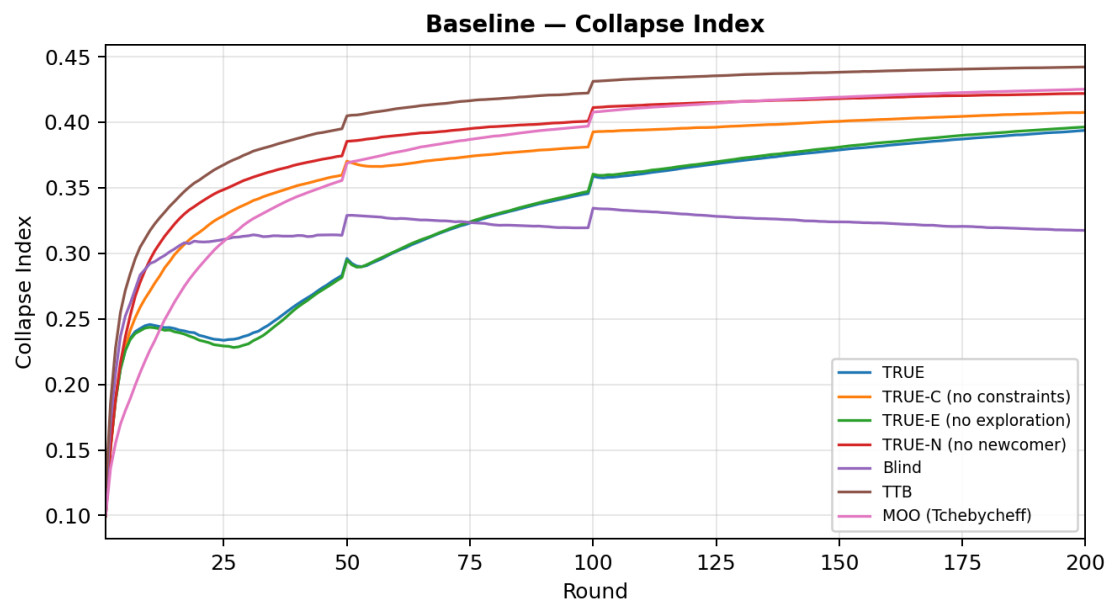
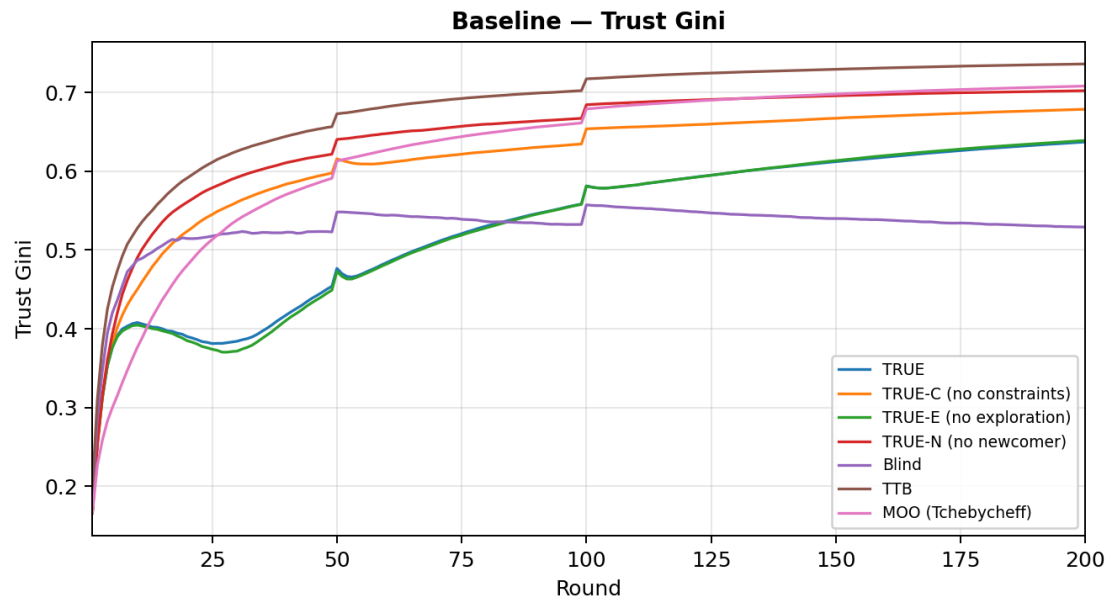
Results Summary

Mean cumulative utility, 95% bootstrap CI, fatal errors, and key metrics per scenario and group.

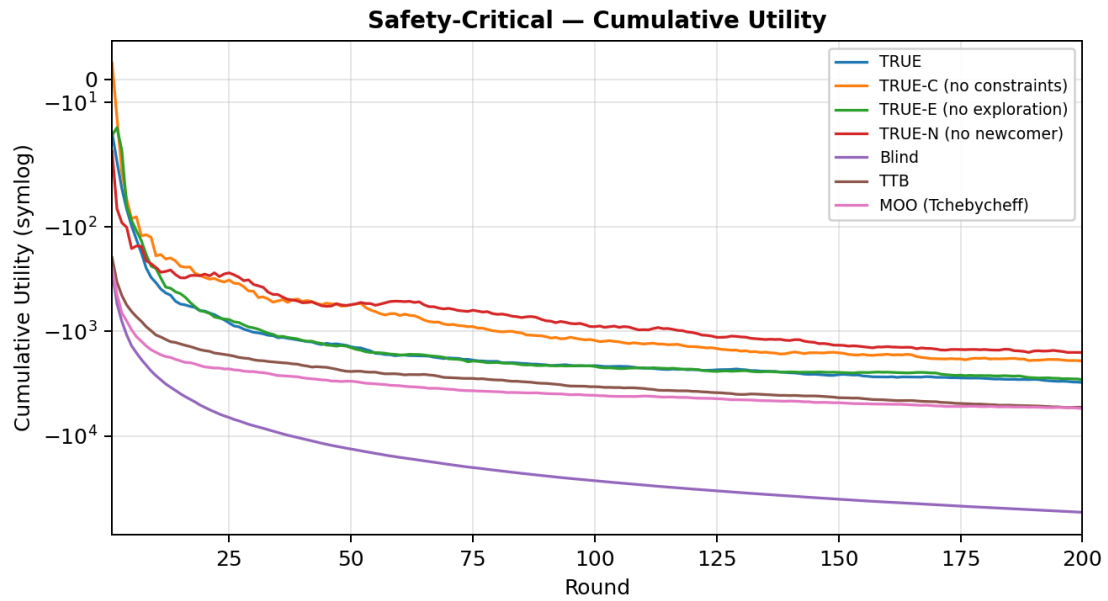
Scenario	Group	Cum.Utility	95% CI	Fatal	True Q	Gini	Collapse	A9 Delay
Baseline	TRUE	896.8	[520.4, 1274.0]	61.0	0.767	0.637	0.394	0.2
Baseline	TRUE-C (no constraints)	1377.6	[1037.2, 1739.0]	60.7	0.799	0.679	0.408	0.9
Baseline	TRUE-E (no exploration)	779.2	[451.0, 1103.0]	61.0	0.765	0.639	0.396	0.2
Baseline	TRUE-N (no newcomer)	1326.0	[974.4, 1671.8]	60.5	0.798	0.702	0.422	383.9
Baseline	Blind	-23753.6	[-24089.4, -23428.6]	208.2	0.343	0.529	0.317	6.8
Baseline	TTB	174.8	[-289.0, 599.6]	64.7	0.764	0.736	0.442	859.7
Baseline	MOO (Tchebycheff)	-547.4	[-1097.4, -40.4]	69.2	0.769	0.708	0.425	789.9
Safety-Critical	TRUE	-3079.0	[-3677.6, -2488.6]	40.4	0.797	0.661	0.418	0.5
Safety-Critical	TRUE-C (no constraints)	-1918.0	[-2516.0, -1328.2]	39.1	0.831	0.688	0.420	1.1
Safety-Critical	TRUE-E (no exploration)	-2895.4	[-3402.0, -2418.0]	39.8	0.799	0.657	0.414	0.6
Safety-Critical	TRUE-N (no newcomer)	-1590.6	[-2162.6, -1009.8]	38.5	0.830	0.723	0.439	890.2
Safety-Critical	Blind	-53661.6	[-54214.6, -53149.4]	213.7	0.347	0.541	0.324	7.7
Safety-Critical	TTB	-5326.2	[-6808.2, -4138.0]	45.9	0.789	0.746	0.454	780.9
Safety-Critical	MOO (Tchebycheff)	-5473.4	[-6306.2, -4684.4]	47.1	0.796	0.713	0.434	809.9
Observation-Manipulated	TRUE	2342.4	[1975.2, 2704.8]	50.4	0.772	0.606	0.365	0.0
Observation-Manipulated	TRUE-C (no constraints)	2600.4	[2256.0, 2933.6]	51.1	0.806	0.657	0.394	0.7
Observation-Manipulated	TRUE-E (no exploration)	2111.2	[1774.0, 2455.2]	51.8	0.769	0.612	0.368	0.0
Observation-Manipulated	TRUE-N (no newcomer)	2894.0	[2521.6, 3264.0]	50.5	0.810	0.664	0.399	6.0
Observation-Manipulated	Blind	-25392.8	[-25722.4, -25050.4]	198.6	0.336	0.517	0.310	7.5
Observation-Manipulated	TTB	359.6	[-676.0, 1235.2]	58.4	0.754	0.739	0.444	860.6
Observation-Manipulated	MOO (Tchebycheff)	-856.8	[-1423.6, -334.4]	65.9	0.753	0.635	0.381	517.5
Utility-Trust-Misalignment	TRUE	2345.4	[2040.2, 2640.5]	56.1	0.749	0.619	0.381	0.1
Utility-Trust-Misalignment	TRUE-C (no constraints)	2954.1	[2640.3, 3284.8]	55.6	0.782	0.659	0.396	0.7
Utility-Trust-Misalignment	TRUE-E (no exploration)	2473.9	[2188.1, 2762.3]	54.9	0.748	0.621	0.384	0.3
Utility-Trust-Misalignment	TRUE-N (no newcomer)	3332.8	[3057.4, 3626.7]	54.2	0.784	0.679	0.408	407.3
Utility-Trust-Misalignment	Blind	-21939.2	[-22229.4, -21643.8]	206.9	0.335	0.533	0.320	7.3
Utility-Trust-Misalignment	TTB	934.2	[74.9, 1645.6]	62.4	0.741	0.733	0.440	799.8
Utility-Trust-Misalignment	MOO (Tchebycheff)	1442.8	[865.3, 1954.9]	62.3	0.751	0.700	0.420	771.4

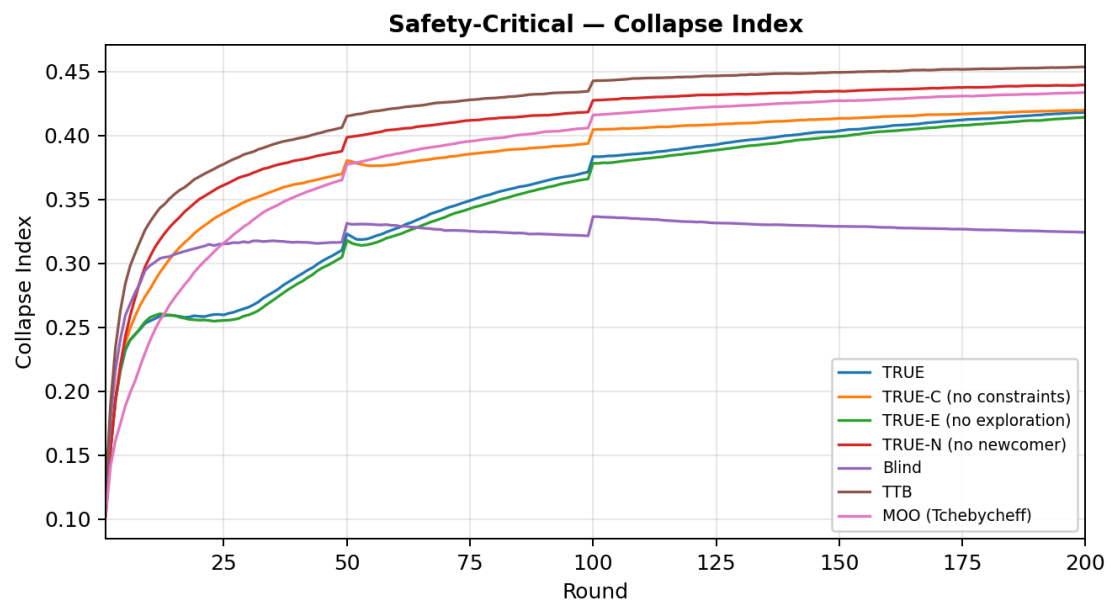
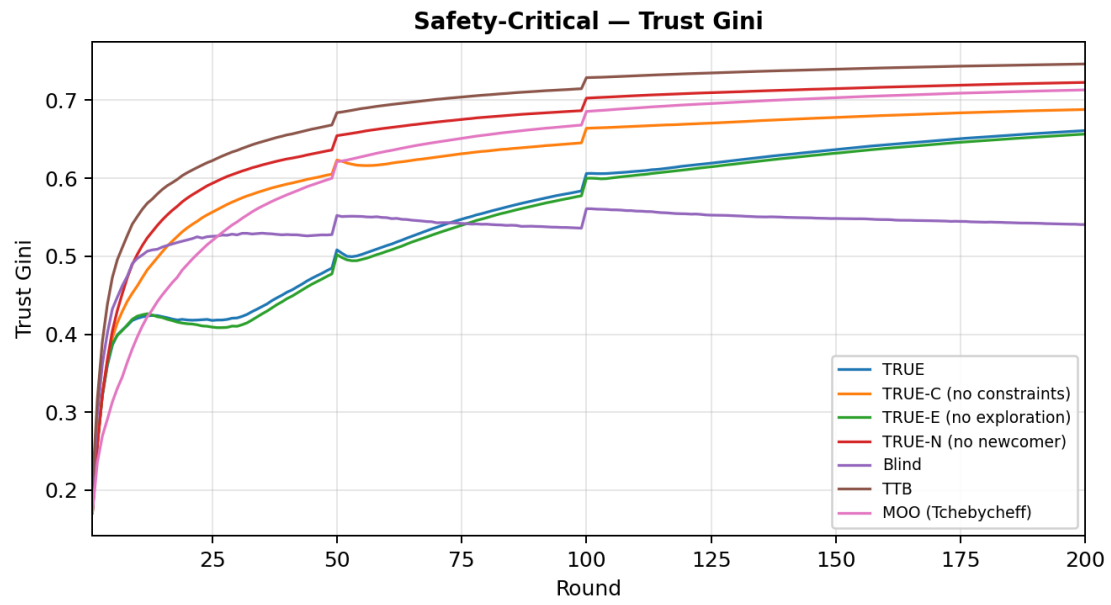
Baseline — Time-Series Analysis



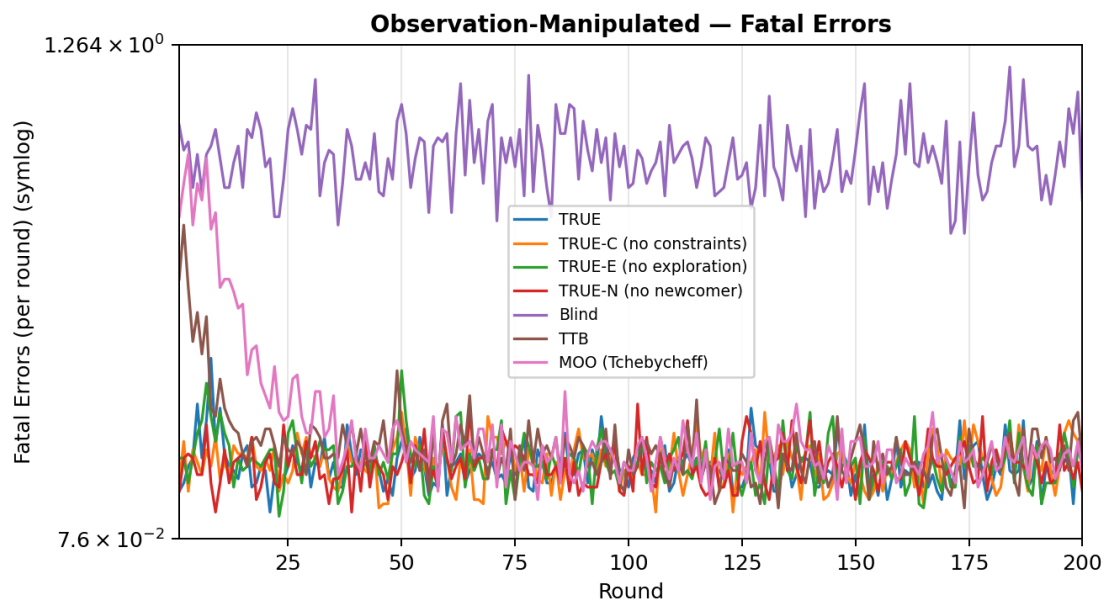
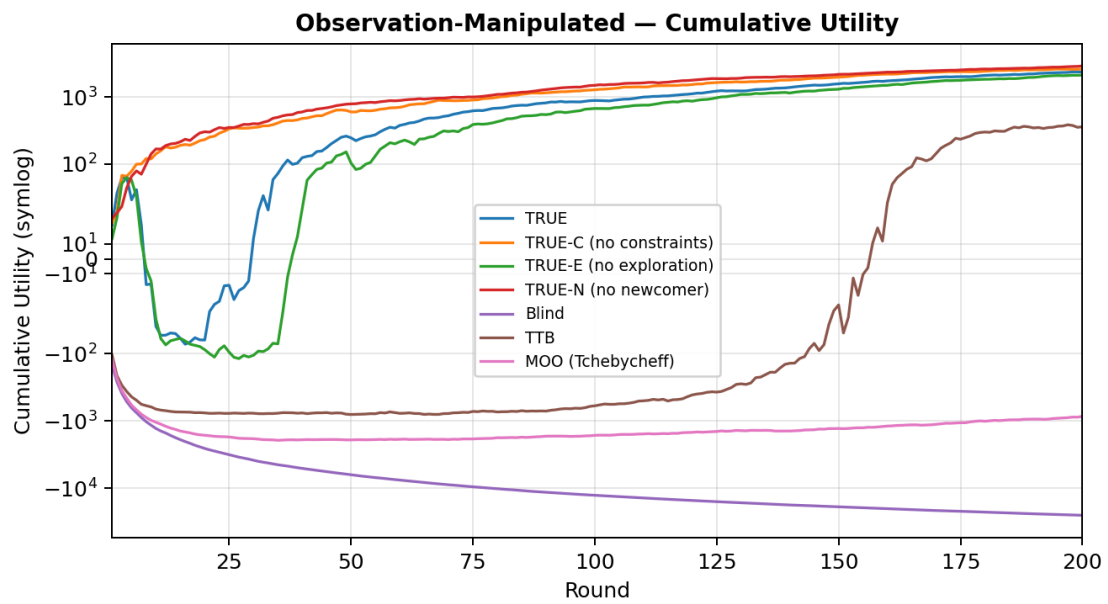


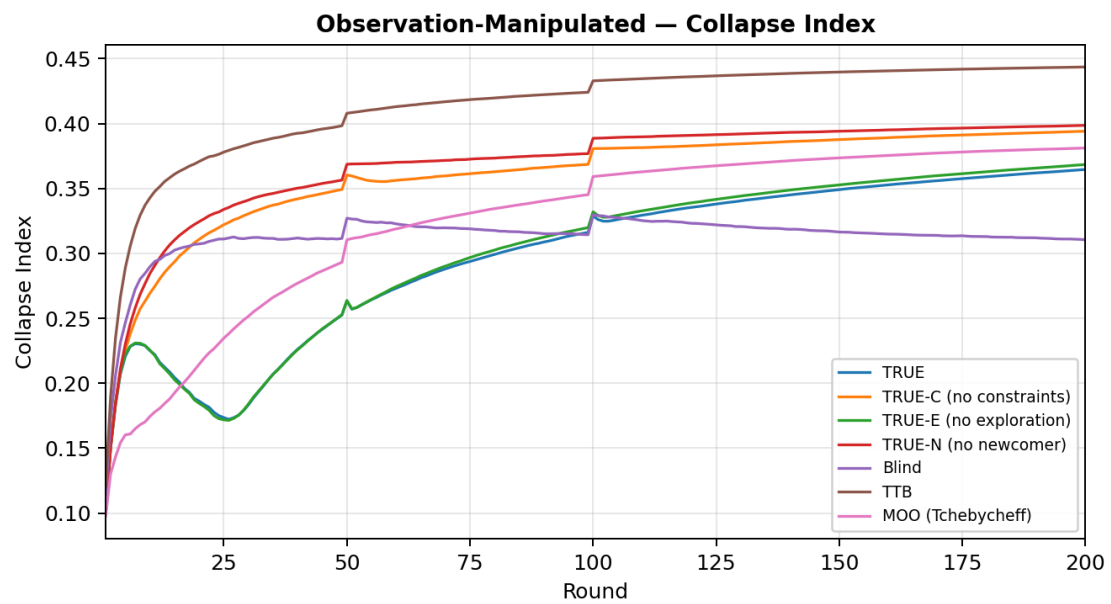
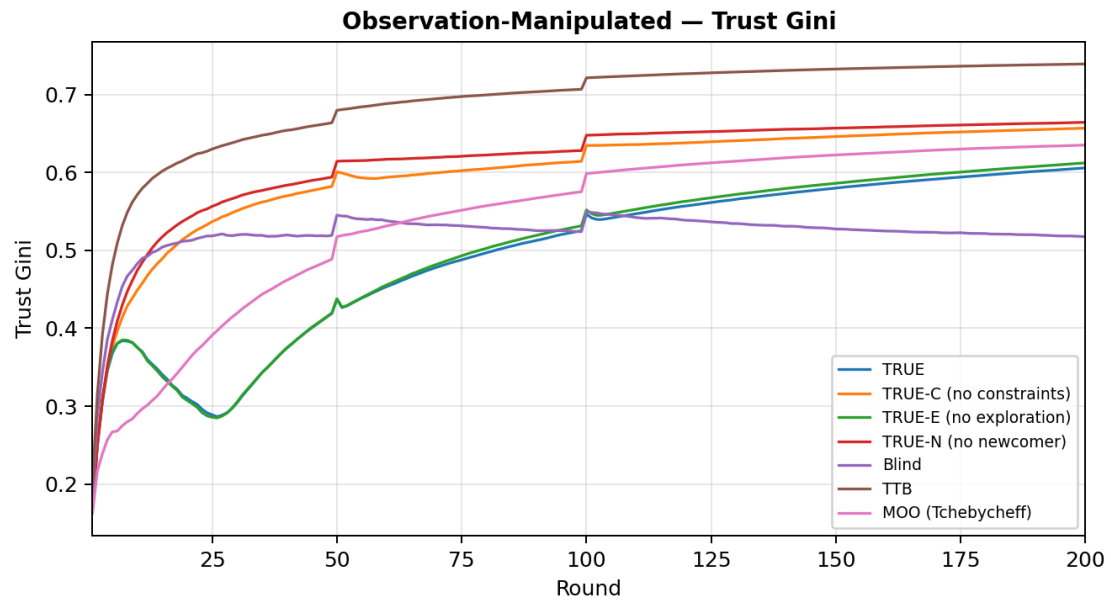
Safety-Critical — Time-Series Analysis



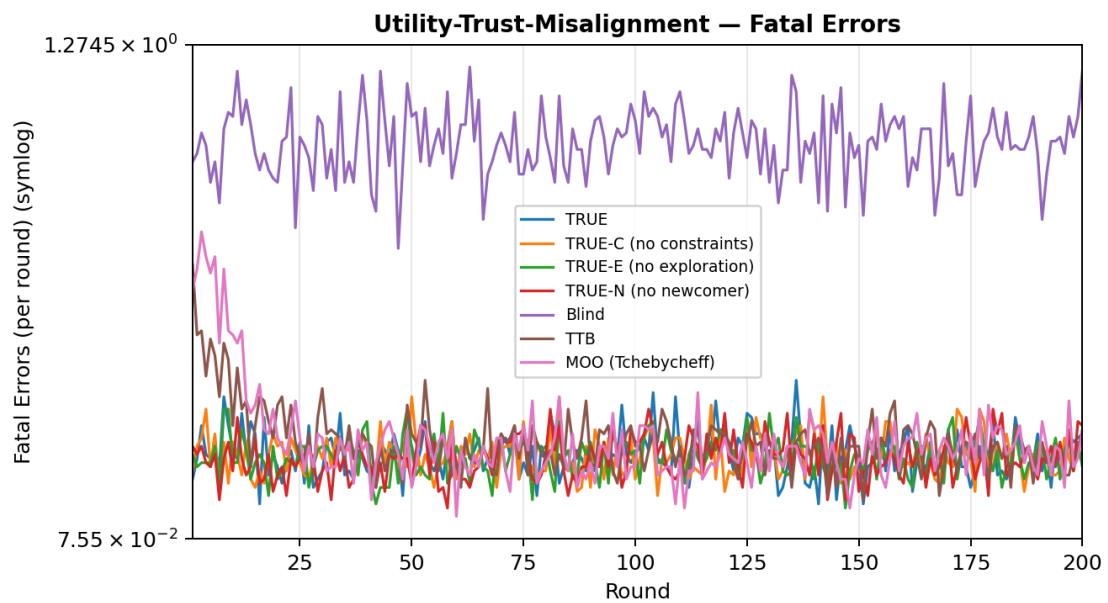
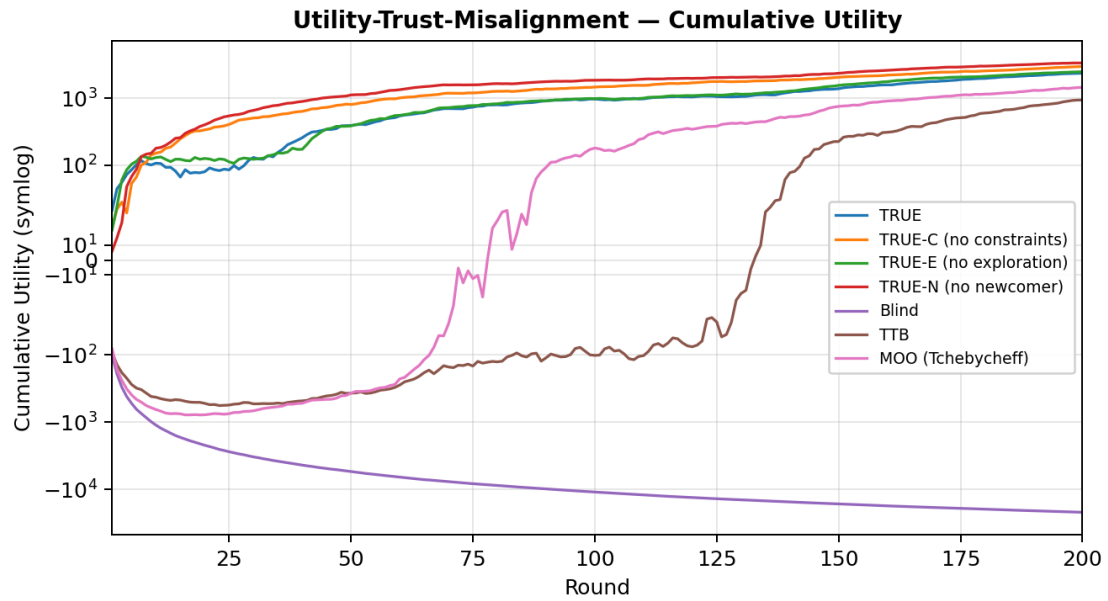


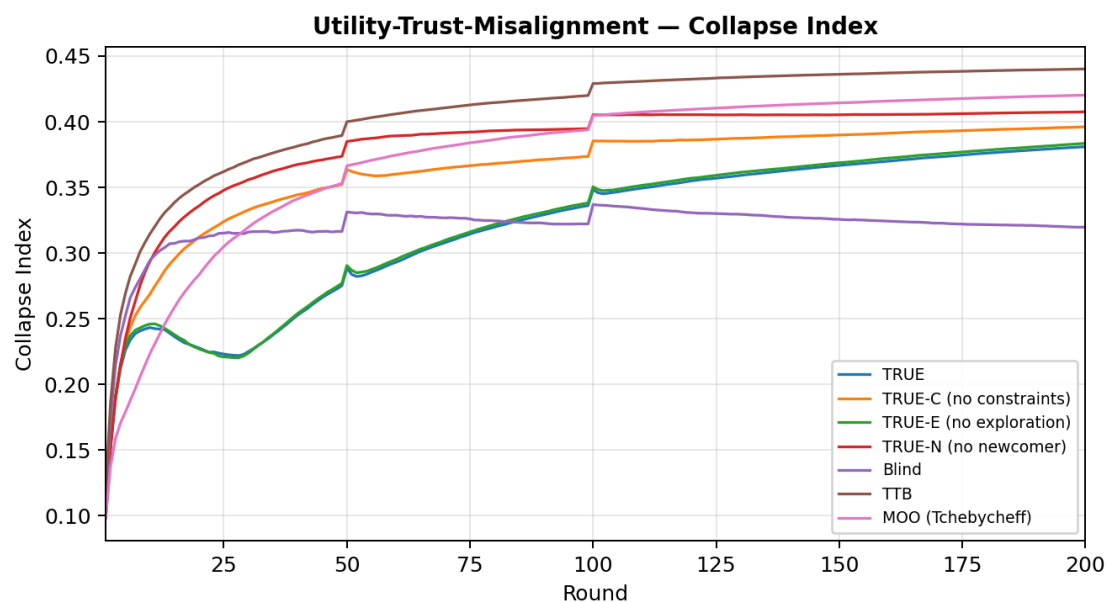
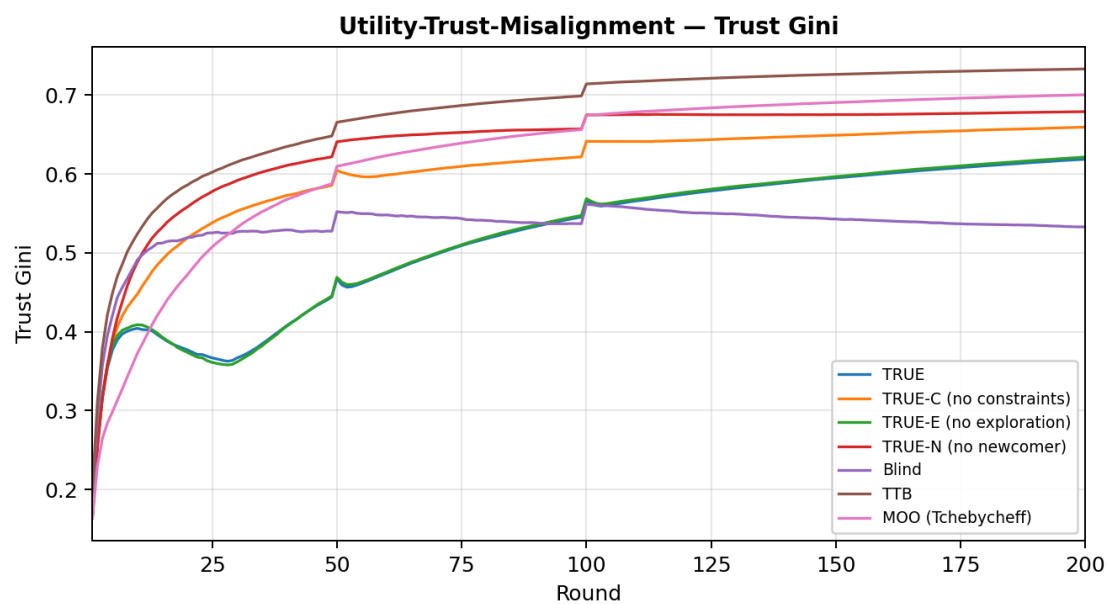
Observation-Manipulated — Time-Series Analysis





Utility-Trust-Misalignment — Time-Series Analysis





Hypothesis Tests (Selected)

All p-values reported with Bonferroni correction across the full test battery.

Scenario	Hypothesis	Mean Diff	t	p_t(Bonf)	Cohen d
Baseline	TRUE_U > Blind_U	24650.4	116.80	0.0000	11.7
Baseline	TRUE_Fatal < Blind_Fatal	147.2	121.14	0.0000	12.1
Baseline	TRUE_U > TTB_U	722.0	2.48	1.0000	0.2
Baseline	TRUE_Fatal < TTB_Fatal	3.7	2.78	0.6976	0.3
Baseline	TRUE_U > MOO_U	1444.2	4.31	0.0020	0.4
Baseline	TRUE_Fatal < MOO_Fatal	8.2	5.13	0.0000	0.5
Baseline	TRUE_Collapse < TTB_Collapse	0.0	21.07	0.0000	2.1
Baseline	TRUE_A9 < Blind_A9	6.5	9.15	0.0000	0.9
Observation-Manipulated	TRUE_U > Blind_U	27735.2	125.99	0.0000	12.6
Observation-Manipulated	TRUE_Fatal < Blind_Fatal	148.2	123.00	0.0000	12.3
Observation-Manipulated	TRUE_U > TTB_U	1982.8	3.92	0.0112	0.4
Observation-Manipulated	TRUE_Fatal < TTB_Fatal	8.0	3.39	0.0891	0.3
Observation-Manipulated	TRUE_U > MOO_U	3199.2	9.85	0.0000	1.0
Observation-Manipulated	TRUE_Fatal < MOO_Fatal	15.5	10.51	0.0000	1.1
Observation-Manipulated	TRUE_Collapse < TTB_Collapse	0.1	46.62	0.0000	4.7
Observation-Manipulated	TRUE_A9 < Blind_A9	7.5	10.59	0.0000	1.1
Safety-Critical	TRUE_U > Blind_U	50582.6	153.22	0.0000	15.3
Safety-Critical	TRUE_Fatal < Blind_Fatal	173.2	134.29	0.0000	13.4
Safety-Critical	TRUE_U > TTB_U	2247.2	2.90	0.4810	0.3
Safety-Critical	TRUE_Fatal < TTB_Fatal	5.5	2.81	0.6319	0.3
Safety-Critical	TRUE_U > MOO_U	2394.4	4.87	0.0001	0.5
Safety-Critical	TRUE_Fatal < MOO_Fatal	6.7	4.94	0.0001	0.5
Safety-Critical	TRUE_Collapse < TTB_Collapse	0.0	17.39	0.0000	1.7
Safety-Critical	TRUE_A9 < Blind_A9	7.2	9.61	0.0000	1.0
Utility-Trust-Misalignment	TRUE_U > Blind_U	24284.6	118.42	0.0000	11.8
Utility-Trust-Misalignment	TRUE_Fatal < Blind_Fatal	150.8	107.40	0.0000	10.7
Utility-Trust-Misalignment	TRUE_U > TTB_U	1411.2	3.32	0.1158	0.3
Utility-Trust-Misalignment	TRUE_Fatal < TTB_Fatal	6.3	3.08	0.2612	0.3
Utility-Trust-Misalignment	TRUE_U > MOO_U	902.6	2.97	0.3794	0.3
Utility-Trust-Misalignment	TRUE_Fatal < MOO_Fatal	6.2	4.21	0.0033	0.4
Utility-Trust-Misalignment	TRUE_Collapse < TTB_Collapse	0.1	23.21	0.0000	2.3
Utility-Trust-Misalignment	TRUE_A9 < Blind_A9	7.2	9.64	0.0000	1.0

Interpretation & Limitations

Task-flow pairing: Within each run all groups see the same module sequence. Differences are therefore attributable to selection mechanisms, not task luck.

Observation uniformity: A8 surface-quality bonus is uniform across groups. The only A8 asymmetry in TTB is its scoring-function bonus (hard-coded trust-objective weight), which is a mechanism-level difference. MOO has no hard-coded A8 bonus; any A8 advantage emerges naturally from high surface-quality observations.

MOO vs TTB: The weighted Tchebycheff MOO baseline includes explicit diversity objective (selection concentration penalty) and per-candidate-pool normalization. It consistently outperforms TTB but still falls short of TRUE, confirming that the problem is "trust-as-objective" rather than the specific aggregation function.

Ablation surprise: TRUE-C and TRUE-N outperform full TRUE in cumulative utility under current parameters, suggesting that constraint filtering and newcomer protection may be overly conservative. However, TRUE-N shows dramatically higher A9 cold-start delay, confirming that newcomer protection serves its intended purpose.

Bonferroni correction: Conservative; hypotheses remaining significant after correction are robust.